

Koopman Modes & Eigenpairs

Consider a discrete-time dynamical system:

$$(DS1) \quad X_{k+1} = \underbrace{F(X_k)}_{F: \Omega \rightarrow \Omega}, \quad k=0,1,2,3,\dots$$

The Koopman operator acts on observables:

$$[Kg](x) = [g \circ \underbrace{F}_{\text{observable}}](x), \quad g: \Omega \rightarrow \mathbb{C}.$$

Key Idea: Study the linear dynamics of observables, which follow the discrete-time law:

$$(DS2) \quad g_{k+1} = Kg_k, \quad k=0,1,2,\dots$$

If $Kv_j = \lambda_j v_j$ for $j=1,\dots,n$, and we have

$$g_0(x) = c_1 v_1(x) + \dots + c_n v_n(x),$$

$$\text{then, } g_{k+1}(x) = [K^{k+1} g_0](x) = \sum_{j=1}^n c_j \lambda_j^{k+1} v_j(x).$$

Koopman
Mode
Decomp.

Q: What does the Koopman Mode Decomp. tell us about the original dynamics (DS1)?

Koopman Eigenfunctions

The Koopman eigenfunctions encode geometric & dynamic information about the state-space of (DS1).

Example. Linear Dynamics

$$x_{k+1} = Ax_k, \quad k = 0, 1, 2, \dots$$

$\Rightarrow$ Solutions to $Av = \lambda v$ define invariant directions in the state-space.

$\Rightarrow$ Solutions to $A^*w = \lambda^*w$ define invariant directions in the space of observables.

$$K(\underbrace{w^*x}) = w^*Ax = \lambda(\underbrace{w^*x})$$

$\phi(x) = \text{eigenfunction}$

If A is diagonalizable then the "left" and "right" eigenvectors are biorthogonal:

$$AV = V\Lambda \Rightarrow V^{-1}A = \Lambda V^{-1} \Rightarrow W = V^{-1}$$

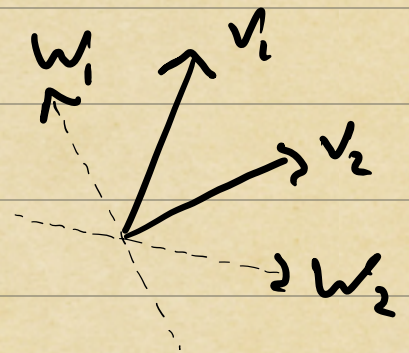
$$\Rightarrow VW = WV = I.$$

This means that the w_i^*x satisfies

$$w_i^*(c_1v_1 + \dots + c_nv_n) = c_i.$$

$\Rightarrow$ Zero level sets of these linear Koopman eigenfunctions encode the invariant subspaces in state space.

$\Rightarrow$ The eigen w_i^*x grows linearly away from this invariant subspace.



$\Rightarrow$ Level sets of w_i^*x are hyperplanes oriented parallel to invariant subspaces.

"Lattice" Structure of Eigenpairs

In general, the eigenvalues of Koopman operators form a lattice in $\mathbb{C}$. If

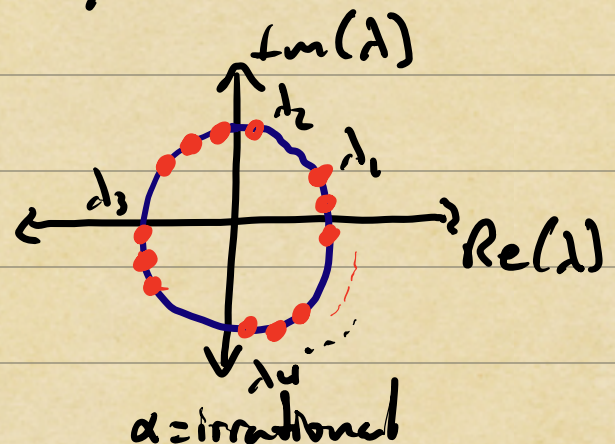
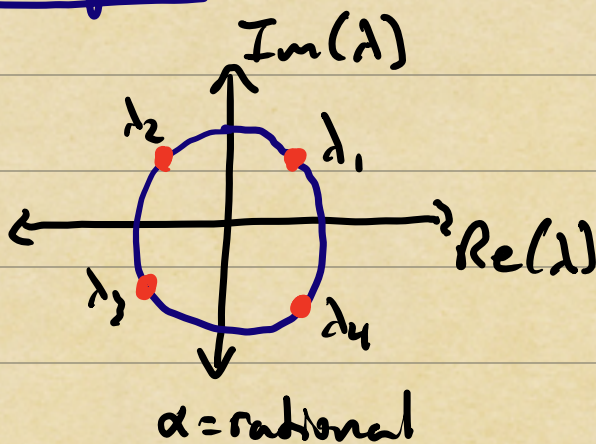
$$Kv_1 = \lambda_1 v_1 \quad \text{and} \quad Kv_2 = \lambda_2 v_2,$$

$$\Rightarrow K(v_1(x)v_2(x)) = v_1(F(x))v_2(F(x))$$

$$= [Kv_1](x)[Kv_2](x) = \lambda_1 \lambda_2 v_1(x)v_2(x).$$

This can be used to generate more Koopman eigenpairs, once a few have been found.

Example. If K is unitary, then $|\lambda| = 1$.



$$\lambda_1 = e^{2\pi i \alpha} \quad \Rightarrow \quad \lambda_j = \lambda_1^j = e^{2\pi i j \alpha}$$

$\Rightarrow$ Eigenvalues may be dense in $\{|z|=1\}$!

Level Sets of Eigenpairs

Beyond linear dynamics, level sets associated w/ Koopman eigenfunctions continue to encode important geometric/dynamic state-space info.

Suppose that $[Kv](x) = \lambda v(x)$, and write $v(x)$ and λ in polar form, so that

$$v(x) = \underbrace{R(x)}_{\text{modulus}} e^{i \underbrace{\Theta(x)}_{\text{argument}}}, \quad \lambda = r e^{i\theta}$$

Consider the sublevel sets defined by

$$M_v(C) = \{x \in \Omega \mid R(x) \leq C\}$$

$$A_v(\alpha) = \{x \in \Omega \mid \Theta(x) \leq \alpha\}$$

Q: How does $x \in M_v(C)$ or $\in A_v(\alpha)$ evolve?

Suppose that $x \in M_v(C)$ and compute

$$v(F(x)) = [Kv](x) = \lambda v(x) = rR(x)e^{i(\theta + \Theta(x))}.$$

Therefore, $x \in M_v(C)$, then $F(x) \in M_v(rC)$.
Similarly, if $x \in A_v(\alpha)$, then $F(x) \in A_v(\alpha + \Theta)$.

These observations allow us to use sub-level sets of the modulus! argument of Rospman eigenfunctions to study basins of attraction, recurrence! periodicity, and more.